$$\chi_0(\vec{q},\omega) = \frac{1}{V}\sum_k \frac{f(\xi_k) - f(\xi_{k+q})}{\omega + \xi_k - \xi_{k+q} + i\delta}$$

where $\xi_k = \varepsilon(\vec{k}) - \mu$.

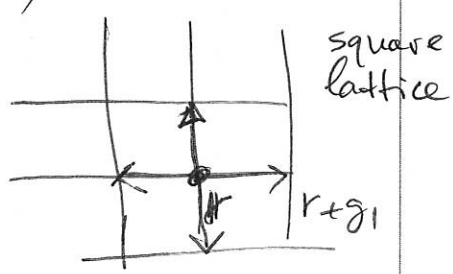
So far we have considered simplest free-electron dispersion

$$\varepsilon_k = \frac{\hbar^2 \vec{k}^2}{2m}.$$

Consider now, as an example, square lattice

$$H_{kin} = -t\sum_{rg}\left(c^+_{r\sigma} c_{r+g\sigma} + h.c.\right)$$

$$= -t\sum_{rg}\frac{1}{N}\sum_{k_1,k_2}\left(e^{-i\bar{k}_1\bar{r}} e^{i\bar{k}_2(\bar{r}+\bar{g})} \cdot c^+_{k_1} c_{k_2} + h.c.\right)$$



$$= -t\sum_{\vec{k}\vec{g}}\left(e^{i\vec{k}\cdot\vec{g}} c^+_{k\sigma} c_{k\sigma} + h.c.\right) =$$

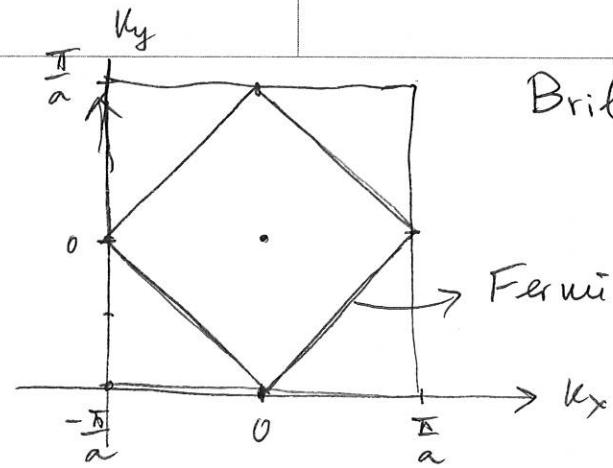
$$= -t\sum_k\left(\sum_{\vec{g}} 2\cos\vec{k}\cdot\vec{g}\right) c^+_{k\sigma} c_{k\sigma} = \sum_{\vec{k}\,\sigma} \varepsilon(k)\, c^+_{k\sigma} c_{k\sigma}$$

$$\varepsilon(\vec{k}) = -4t\left(\cos(k_x a) + \cos(k_y a)\right)$$

Clearly for $\vec{k} = (k_x, k_y) \simeq 0$ we can expand cosines

$$\varepsilon(\vec{k}\sim 0) = -4t\left(1 - \tfrac{1}{2}(k_x a)^2 + 1 - \tfrac{1}{2}(k_y a)^2\right) =$$

$$= -8t + 2ta^2(k_x^2 + k_y^2)$$



Brillouin Zone

(1 electron per site)

Fermi surface of half-filled Hubbard model

Find $\vec{k}$ for which $\varepsilon(\vec{k}) = 0$:

$$\cos(k_x a) + \cos(k_y a) = 0 \Rightarrow \begin{cases} k_y a = \pi - k_x a \\ \text{or} \\ k_y a = -\pi + k_x a, \text{ etc.} \end{cases}$$

straight lines in BZ

$$\left.\begin{aligned} \text{Volume enclosed} &= \left(\frac{\sqrt{2}\,\pi}{a}\right)^2 = \frac{2\pi^2}{a^2} \\ \text{Volume of BZ} &= \left(\frac{2\pi}{a}\right)^2 = \frac{4\pi^2}{a^2} \end{aligned}\right\} \frac{1}{2}\text{-filled band}$$

$E_F = \underline{\mu = 0}$ chemical potential.

Observe that $\varepsilon(\vec{k} + \vec{Q}) = -4t\left(\cos(k_x a + \pi) + \cos(k_y a + \pi)\right) =$

$\vec{Q} = \frac{\pi}{a}(1,1)$

$$= 4t\left(\cos k_x a + \cos k_y a\right)$$

$$= -\varepsilon(\vec{k})$$

Thus $\chi_0(\vec{Q}, \omega = 0) = \frac{1}{V} \sum_k \frac{f(\xi_k) - f(\xi_{k+Q})}{\xi_k - \xi_{k+Q}} =$

$$= \frac{1}{V} \sum_k \frac{f(\xi_k) - f(-\xi_k)}{\xi_k - (-\xi_k)}$$

$$f(\xi) = \frac{1}{e^{\frac{\xi}{T}}+1} \quad ; \quad f(-\xi) = \frac{1}{e^{-\xi/T}+1} = \frac{e^{\xi/T}}{e^{\xi/T}+1}$$

$$\chi_0(\vec{Q},0) = \int d\xi \, \underbrace{N(\xi)}_{\text{density of states}} \, \frac{1-e^{\xi/T}}{e^{\xi/T}+1} \, \frac{1}{2\xi}$$

$$= -\int d\xi \, N(\xi) \, \frac{e^{\frac{\xi}{2T}} - e^{-\frac{\xi}{2T}}}{e^{\frac{\xi}{2T}} + e^{-\frac{\xi}{2T}}} \, \frac{1}{2\xi} =$$

$$= -\int_{-\lambda}^{\lambda} d\xi \, N(\xi) \, \frac{\tanh\left(\frac{\xi}{2T}\right)}{2\xi} \approx -N(0) \int_0^{\lambda} d\xi \, \frac{\tanh\left(\frac{\xi}{2T}\right)}{\xi}$$

λ is of the order of bandwidth.

$$\int_0^{\lambda} d\xi \, \frac{\tanh\left(\frac{\xi}{2T}\right)}{\xi} = \left(\int_0^{c} d\xi + \int_c^{\lambda} d\xi\right) \frac{\tanh\left(\frac{\xi}{2T}\right)}{\xi}$$

$x = \frac{\xi}{2T}$

c is such that $\tanh\left(\frac{c}{2T}\right) \approx \frac{c}{2T} \ll 1$

so that in the interval $\xi \in (0, c)$ we can approximate

$\tanh\left(\frac{\xi}{2T}\right) \approx \frac{\xi}{2T}$

$$\Rightarrow \int_0^{c} d\xi \, \frac{\xi}{2T} \frac{1}{\xi} = \frac{c}{2T} \ll 1.$$

In the other interval $\xi \in (c, \lambda)$ $\tanh\left(\frac{\xi}{2T}\right) \approx 1$

$\frac{\xi}{2T} \gg 1.$

let $x = \frac{\xi}{2T} \Rightarrow \xi = 2Tx$, $d\xi = 2T\,dx$

$$\int_0^{\lambda} d\xi \, \frac{th\left(\frac{\xi}{2T}\right)}{\xi} = \int_0^{\lambda/2T} \frac{dx}{x} \, th(x) = \int_0^1 \frac{dx}{x} \, th(x) +$$

$$+ \int_1^{\lambda/2T} \frac{dx}{x} \, th(x) \approx \int_0^1 dx + \int_1^{\lambda/2T} \frac{dx}{x} \approx \ln\left(\frac{C\lambda}{2T}\right)$$

where $C \sim O(1)$ constant.

$$\Rightarrow \chi_0(\vec{Q}, 0) = -N(0) \ln\left(\frac{C\lambda}{T}\right)$$

diverges logarithmically as $T \to 0$.

Now, $\chi_c(\vec{Q}, \omega=0) = \dfrac{\chi_0(\vec{Q},0)}{1 - U\chi_0(\vec{Q},0)}$

charge

$$\chi_s(\vec{Q}, \omega=0) = \frac{\chi_0(\vec{Q},0)}{1 + U\chi_0(\vec{Q},0)}$$

spin

$$\cancel{\chi_c = \frac{1}{\chi_0^{-1} - U} = \ldots}$$

Instability if denominator $= 0$

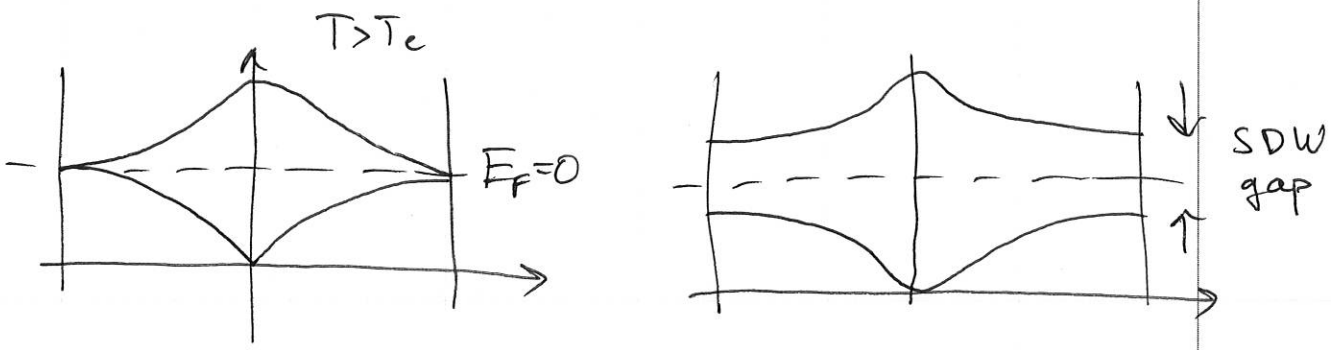
spin channel $\quad 1 + U\left(-N_0 \ln\left(\frac{C\lambda}{T_0}\right)\right) = 0 \Rightarrow \ln\left(\frac{C\lambda}{T_0}\right) = \frac{1}{N_0 U}$

$$T_0 = c\lambda\, e^{-1/N_0 U}$$ ← below this T_0 we have ~~the spin~~ spin-density wave

charge channel: $1 - U\left(-N_0 \ln\left(\frac{c\lambda}{T}\right)\right) =$

$$= 1 + U N_0 \ln\left(\frac{c\lambda}{T}\right) > 1 \text{ always}$$

⇒ no instability



The assumption that density of states is <u>smooth</u> is a standard one in the theory of metals. Nonetheless it is not always true.
Let us check it here:

$$N(\varepsilon) = \int \frac{d^2\vec{k}}{(2\pi)^2}\, \delta(\varepsilon - \varepsilon(k))$$

where $\varepsilon(k) = -4t(\cos k_x a + \cos k_y a)$ is the particle's dispersion.
Suppose we are interested in energies near the bottom of the band, i.e. $\varepsilon \approx -8t$.
This means $\vec{k} \approx 0$ and we can expand:

$$\varepsilon(\vec{k}\sim 0) \approx -4t\left(2 - \frac{a^2}{2}(k_x^2+k_y^2)\right) =$$

$$= -8t + \underbrace{2ta^2(k_x^2+k_y^2)}_{\text{free particle with mass } m = \frac{1}{4ta^2}} = -\varepsilon_0 + \frac{\vec{k}^2}{2m}$$

Then $N(\varepsilon) = \int \frac{dk\cdot k}{(2\pi)^2}\int_0^{2\pi} d\varphi\, \delta\left(\varepsilon - \left(\varepsilon_0 + \frac{k^2}{2m}\right)\right) =$

$$= \int \frac{d(k^2)}{4\pi}\, \delta\left(\varepsilon - \varepsilon_0 - \frac{k^2}{2m}\right) = \left(x = \frac{k^2}{2m}\right) =$$

$$= \frac{2m}{4\pi}\int dx\, \delta(\varepsilon - \varepsilon_0 - x) = \begin{cases} \frac{m}{2\pi} & \text{if } \varepsilon > \varepsilon_0 \\ 0 & \text{if } \varepsilon < \varepsilon_0 \end{cases}$$

So for energies near the band edge we get usual energy-independent DoS in two dimensions.

However this is not true for all energies: consider $\varepsilon \approx 0$, i.e. energies in the middle of the band. There we have $\cos(k_x a) \approx -\cos(k_y a)$ and, most importantly, 4 points where $k_x = 0 \quad \pm\frac{\pi}{a}$

$k_y = \pm\frac{\pi}{a} \quad 0$

Consider one such point: $k_x \approx \frac{\pi}{a}$, $k_y \approx 0$

$$\Rightarrow \cos(k_x a) \approx -\left(1 - \frac{(q_x a)^2}{2}\right) \quad \text{where } q_x = k_x - \pi$$

$$\cos(k_y a) \approx 1 - \frac{1}{2}(k_y a)^2.$$

$$\varepsilon(\vec{k}) \approx -4t\left(-1 + \frac{1}{2} q_x^2 a^2 + 1 - \frac{1}{2} k_y^2 a^2\right) =$$

$$= -2ta^2\left(q_x^2 - k_y^2\right).$$

From this range of momenta we get

$$N(\varepsilon) = 4\int \frac{dq_x\, dk_y}{(2\pi)^2}\, \delta\left(\varepsilon - \frac{k_y^2 - q_x^2}{2m}\right) =$$

four corners

$$= 4\int \frac{dq_x}{(2\pi)^2} \int dk_y\, \delta\left(\varepsilon + \frac{q_x^2}{2m} - \frac{k_y^2}{2m}\right)$$

Use $\int dx\, \delta(f(x)) = \int dx \sum_{x_0} \frac{\delta(x - x_0)}{|f'(x_0)|} = \sum_{x_0} \frac{1}{|f'(x_0)|}$.

where $f(x_0) = 0$.

Here we have $f(k_y) = \varepsilon + \frac{q_x^2}{2m} - \frac{k_y^2}{2m} \Rightarrow (k_y)_0 = \pm\sqrt{2m\varepsilon + q_x^2}$

$$f'(k_{y0}) = -\frac{k_{y0}}{m} = \mp\frac{\sqrt{2m\varepsilon + q_x^2}}{m}$$

$$\Rightarrow N(\varepsilon) = \frac{4}{(2\pi)^2}\int_0^{\Lambda} dq_x \, \frac{2m}{\sqrt{2m\varepsilon + q_x^2}} = \frac{2m}{\pi^2} \ln\left|\sqrt{2m\varepsilon + q_x^2} + q_x\right| \Big|_0^{\Lambda}$$

$$= \frac{2m}{\pi^2} \ln \frac{2\Lambda}{\sqrt{2m\varepsilon}} \simeq \frac{m}{\pi^2} \ln\left(\frac{c}{\varepsilon}\right).$$

where constant c absorbs all numerical factors.

Thus, near $\varepsilon = 0$ $N(\varepsilon)$ has logarithmic singularity!

Hence approximation which we made

$$\int_0^{\lambda} d\xi \, N(\xi) \, \frac{\tanh(\xi/2T)}{\xi} \simeq N(0) \int_0^{\lambda} d\xi \, \frac{\tanh(\xi/2T)}{\xi}$$

is not really valid: $N(\xi = 0) = \infty$!

Need to re-do: $\chi_0(\vec{q}, 0) = -\int_0^{\lambda} d\xi \, \frac{m}{\pi^2} \ln\left(\frac{c/2T}{\varepsilon/2T}\right) \frac{\tanh\left(\frac{\varepsilon}{2T}\right)}{\varepsilon}$

$\left(x = \frac{\varepsilon}{2T}\right)$

$$= -\frac{m}{\pi^2} \int_0^{\lambda/2T} dx \left\{ \ln\left(\frac{c}{2T}\right) - \ln(x) \right\} \frac{\tanh(x)}{x}$$

$$\cancel{\simeq \frac{m}{\pi^2} \int_1^{\lambda/2T} dx \, \frac{\ln x}{x} = \frac{m}{\pi^2} \frac{1}{2} (\ln x)^2 = \frac{m}{2\pi^2} \left(\ln \frac{\lambda}{2T}\right)^2}.$$

$$\simeq -\frac{m}{\pi^2} \left\{ \ln\left(\frac{c}{2T}\right) \int_1^{\lambda/2T} \frac{dx}{x} - \int_1^{\lambda/2T} \frac{dx}{x} \ln(x) \right\}$$

$$= -\frac{m}{\pi^2} \left\{ \ln\left(\frac{c}{2T}\right) \ln\left(\frac{\lambda}{2T}\right) - \frac{1}{2} \left(\ln \frac{\lambda}{2T}\right)^2 \right\} =$$

$$= -\frac{m}{\pi^2} \ln\left(\frac{\lambda}{2T}\right) \left\{ \ln\left(\frac{c}{2T}\right) - \frac{1}{2} \ln\left(\frac{\lambda}{2T}\right) \right\} = -\tilde{c} \ln^2\left(\frac{\lambda}{2T}\right)$$

$\Rightarrow$ $\chi_0(\vec{Q})$ diverges as $\left(\ln\frac{1}{T}\right)^2 \gg \ln\frac{1}{T}$.

$\Rightarrow$ instability condition becomes

$$1 + U\left(-\tilde{c}\,\ln^2\left(\frac{\lambda}{2T_1}\right)\right) = 0$$

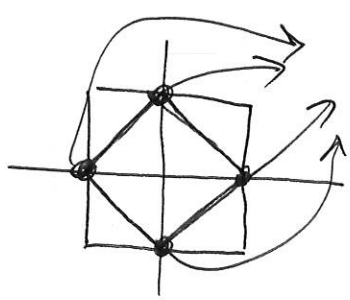
$$\ln^2\left(\frac{\lambda}{2T_1}\right) = \frac{1}{\tilde{c}U} \Rightarrow \ln\left(\frac{\lambda}{2T_1}\right) = \frac{1}{\sqrt{\tilde{c}U}}$$

$$\frac{\lambda}{2T_1} = e^{\frac{1}{(\tilde{c}U)^{1/2}}}$$

$$T_1 = \frac{\lambda}{2}\, e^{-\frac{1}{\sqrt{\tilde{c}U}}}.$$

Important point is that $T_1 \gg T_0$

(because $e^{-\frac{1}{\sqrt{x}}} \gg e^{-1/x}$ for $x \to 0$)



points where Fermi surface touches Brillouin zone are van Hove points where electron dispersion is quadratic!

These points enhance singularity and lead to stronger dependence of instability temperature $T(U)$ on interaction U.